Design of a 6-V Lead-Acid Battery Charger Using a Discrete Series Pass Regulator with Voltage-Feedback Control

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***Abstract*—This work describes the design, simulation, and practical implementation of a discrete series-pass linear voltage regulator developed for charging a 6-V sealed lead-acid (SLA) battery. The proposed regulator employs a Zener-diode reference with a feedback network to ensure a stable charging voltage, along with a transistor-based current limiting stage for overload protection. LTSpice simulations were used to validate the concept, predicting an output of 7.25 V and a current limit near 650 mA. A hardware prototype was then constructed and evaluated under different loading scenarios. Measured results showed strong line regulation (0.72%) and acceptable load regulation (16.3%), confirming suitability for SLA charging. The current limiting circuit was also experimentally verified to trigger at 651 mA, safeguarding the pass transistor during excessive load conditions. Overall, the study demonstrates that discrete linear regulators remain a practical, low-cost solution for battery charging, while also emphasizing considerations related to power loss and thermal performance.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

Lead-acid batteries remain the dominant energy storage solution for automotive, uninterruptible power supply (UPS), and renewable energy systems due to their robustness and cost-effectiveness. However, their longevity is heavily dependent on the charging profile. A 6-V nominal lead-acid battery requires a charging voltage strictly maintained between 6.8 V and 7.8 V to reach full capacity. Deviations outside this window result in critical failure modes: excessive voltage causes electrolyte gassing and water loss, while insufficient voltage leads to sulfation on the plates. Therefore, a precise voltage regulation circuit is not merely an accessory but a fundamental requirement for battery safety and efficiency.

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The evolution of voltage regulation has progressed from simple Zener-based shunt regulators to modern Switching Mode Power Supplies (SMPS). While SMPS architectures offer high power efficiency, they suffer from electromagnetic interference (EMI) and output ripple. In contrast, Linear Regulators, specifically the Series Pass topology, provide superior line regulation and low output noise, making them ideal for sensitive instrumentation and steady charging applications [1]. As detailed in Jaeger and Blalock [1], the Series Pass regulator functions by operating a power transistor in its active linear region, effectively acting as a variable resistor that continuously adjusts to maintain a constant output voltage despite fluctuations in input voltage or load current.

While integrated circuit (IC) regulators like the LM317 are commonplace, they function as "black boxes" that obscure the internal feedback mechanisms. This project focuses on the design and implementation of a **discrete** series pass regulator with a voltage-feedback loop. By constructing the circuit from individual bipolar junction transistors (BJTs) and a Zener reference, we isolate and analyze the critical feedback path where the error signal—derived from the difference between the sampled output and the reference—drives the pass element. The design aims to regulate a variable DC input (10 V – 30 V) to a stable charging voltage of approximately 7.0 V.

A. Goals and Objectives

The primary objective of this project is to design, simulate, and experimentally validate a discrete linear series-pass voltage regulator capable of safely charging a 6-V sealed lead-acid (SLA) battery. The design targets a regulated output of approximately 7.25 V with integrated current limiting to protect both the battery and the pass transistor. Additional ob-

jectives include analyzing regulator performance under varying load and line conditions, evaluating thermal behavior, and comparing simulated results with hardware measurements to assess real-world feasibility.

The remainder of this paper is organized as follows: Section II details the theoretical analysis and component sizing for the regulator circuit. Section III presents the simulation results and experimental validation on hardware. Section IV discusses the performance metrics, including load regulation and efficiency, and Section V concludes the study.

B. Literature Review

Lead-acid battery chargers have traditionally been implemented using linear regulators, switching-mode power supplies (SMPS), or integrated regulator ICs. Early designs relied on Zener-diode shunt regulators and series-pass transistors due to their simplicity and low noise. Jaeger and Blalock [1] describe the linear series-pass topology as a feedback-controlled element that modulates transistor conduction to maintain a stable output voltage, making it suitable for sensitive charging applications.

Modern chargers commonly employ SMPS architectures, which offer high efficiency but introduce drawbacks such as electromagnetic interference (EMI), switching ripple, and increased design complexity. Prior studies in battery charger design have shown that linear regulators, despite their lower efficiency, remain advantageous for SLA charging due to their smooth voltage profile, low ripple, and predictable current behavior. Commercial IC regulators such as the LM317 provide convenience but encapsulate feedback mechanisms inside the device, limiting educational value. By contrast, discrete implementations allow full visibility of the control loop, enabling students and engineers to analyze the exact role of each component, including the Zener reference, feedback divider, error amplifier, and current limiter.

Research on SLA battery charging (e.g., Panasonic battery datasheets and application notes) emphasizes the need for precise regulation between 6.8 V and 7.4 V to avoid sulfation or electrolyte gassing. The design in this project aligns with those requirements and extends prior educational work by providing a complete analytical, simulated, and hardware-validated discrete regulator.

II. Materials and Methods

A. Analytical Analysis

The designed battery charger is based on a discrete Series Pass Regulator topology incorporating a voltage-feedback loop and an active current-limiting stage. The circuit schematic consists of three primary functional blocks: the pass element (Q_1), the error amplifier (Q_2), and the current limiter (Q_3).

1) Voltage Regulation Loop: The output voltage V_{out} is continuously sampled by a resistive divider network formed by R_3 and R_4. The sampled voltage is applied to the base of the error amplifier (Q_2). The emitter of Q_2 is clamped to a precise reference potential using a Zener diode (D_2) with voltage V_z.

For the error amplifier to operate in the active region, the base potential must exceed the emitter potential by the forward voltage drop $V_{BE} \approx 0.7$ V. The equilibrium condition for the regulation loop is derived as follows:

$$V_{B(Q2)} = V_{out}\left(\frac{R_4}{R_3 + R_4}\right) \tag{1}$$

Since $V_{E(Q2)} = V_z$, the condition $V_B = V_E + V_{BE}$ yields:

$$V_{out}\left(\frac{R_4}{R_3 + R_4}\right) = V_z + V_{BE} \tag{2}$$

Solving for the regulated output voltage:

$$V_{out} = (V_z + V_{BE})\left(1 + \frac{R_3}{R_4}\right) \tag{3}$$

This equation demonstrates that V_{out} can be tuned by adjusting the ratio of R_3 to R_4, while the Zener provides stability against input fluctuations.

2) Current Limiting: To prevent damage to the battery or the pass transistor during fault conditions, a current-sensing mechanism is implemented using Q_3 and sense resistor R_6. The base-emitter junction of Q_3 is connected in parallel with R_6. When the load current I_{load} creates a voltage drop across R_6 equal to the turn-on voltage of Q_3 (≈ 0.65 V), Q_3 begins to conduct, shunting drive current away from Q_1 and clamping the output current at:

$$I_{limit} \approx \frac{0.65 \text{ V}}{R_6} \tag{4}$$

B. Component Selection and Sizing

1) Voltage Setting: The design targets a charging voltage suitable for a 6-V lead-acid battery, typically between 7.2 V and 7.4 V. Using a 5.1 V Zener diode ($V_z = 5.1$ V), we selected feedback resistors $R_3 = 1.5$ kΩ and $R_4 = 6$ kΩ. Substituting these values into Eq. 3:

$$V_{out} = (5.1 + 0.7)\left(1 + \frac{1.5}{6}\right) = 5.8 \times 1.25 = 7.25 \text{ V} \tag{5}$$

This value ensures a safe and effective charging potential.

2) Current Limit Sizing: A current limit of approximately 650 mA was chosen to protect the experimental setup. A sense resistor $R_6 = 1\Omega$ was selected, yielding:

$$I_{limit} \approx \frac{0.65 \text{ V}}{1\Omega} = 650 \text{ mA} \tag{6}$$

3) Power Dissipation: The pass transistor Q_1 (TIP31C) operates in the linear region and dissipates power defined by $P_D = (V_{in} - V_{out})I_{load}$. Under the worst-case simulation condition ($V_{in} = 30$ V and max current):

$$P_{D(max)} \approx (30 \text{ V} - 7.25 \text{ V}) \times 0.65 \text{ A} \approx 14.8 \text{ W} \tag{7}$$

Although the TIP31C has a maximum power rating of 40 W, this applies only when the case is kept at 25°C. Without a heatsink, the thermal resistance would lead to failure. Therefore, a heatsink is strictly required for the hardware implementation.

C. Bill of Materials

Table I lists the specific components selected for the final hardware implementation. These components were chosen based on the sizing calculations and availability defined in the project requirements.

TABLE I
BILL OF MATERIALS

Ref	Component	Value/Rating	Description
Q_1	Pass Transistor	TIP31C	NPN Power BJT
Q_2	Error Amp	BC547	NPN Small Signal
Q_3	Current Limiter	2N2222	NPN Gen. Purpose
D_2	Reference	5.1 V	Zener Diode (1W)
D_1	Protection	1N5822	Schottky (3A, 40V)
R_3	Feedback	1.5 kΩ	1/4W Metal Film
R_4	Feedback	6 kΩ	1/4W Metal Film
R_6	Sense	1 Ω	5W Wire-wound
R_1	Bias	2.2 kΩ	1/4W Resistor
C_{in}	Input Filter	100 μF	35V Electrolytic
C_{out}	Output Filter	0.1 μF	Ceramic

D. Circuit Schematic and Simulation

The complete circuit design was rigorously verified using LTSpice simulation software prior to hardware implementation. The simulation schematic, shown in Fig. 1, incorporates the full series pass topology with the voltage feedback network and the current limiting sub-circuit.

Two distinct simulation analyses were performed to validate the design:

1) **DC Operating Point (.op):** An operating point analysis was conducted to calculate the steady-state node voltages and branch currents at a nominal input voltage ($V_{in} = 15$ V). This step was crucial to verify that the transistors (Q_1, Q_2, and Q_3) were properly biased in their required active or cutoff regions and that the Zener diode was operating in breakdown mode.
2) **DC Sweep (.dc):** A DC sweep analysis was performed by varying the input voltage source (V_{in}) linearly from 10 V to 30 V with a step size of 0.1 V. This simulation characterized the line regulation capabilities, confirming that the output voltage (V_{out}) stabilizes at approximately 7.25 V once V_{in} exceeds the dropout threshold.

E. Hardware Implementation Details

The physical realization of the battery charger was carried out on a solderless breadboard, with careful attention paid to component placement, noise minimization, and thermal management.

1) Construction and Interconnection: The circuit was constructed with a focus on separating high-current power paths from low-current signal paths to maintain stability and measurement accuracy.

- **Power Path:** The primary current path, consisting of the input supply, pass transistor (Q_1), sense resistor (R_6), and the output load, was wired using short, direct jumper connections to minimize parasitic resistance and inductance.
- **Feedback Loop Placement:** The voltage divider resistors (R_3, R_4) and the error amplifier (Q_2) were placed in close proximity to the output node. This ensures that the sampled voltage accurately reflects the battery terminal voltage, minimizing steady-state error. The Zener diode (D_2) was grounded directly to the low-noise common return rail.
- **Current Sensing Integration:** The current limiting transistor (Q_3) was wired directly across the sense resistor R_6 ($1\Omega, 5$W). This tight coupling ensures an immediate response to overcurrent conditions by shunting the base drive of Q_1 when the voltage drop across R_6 exceeds approx. 0.65 V.

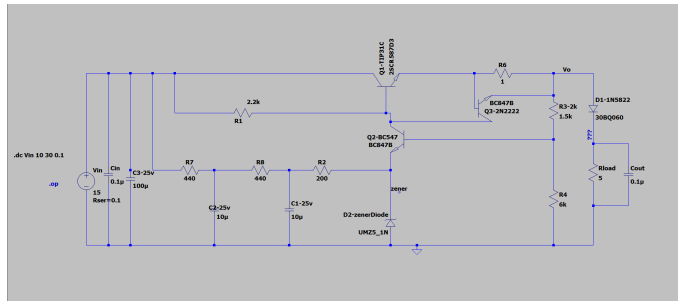

Fig. 1. LTSpice simulation schematic showing both simulation directives: the DC Operating Point (.op) for biasing verification and the DC Sweep (.dc) for regulation analysis.

2) Thermal Management: The linear series pass topology inherently operates by dissipating excess voltage as heat. Under worst-case conditions (Input $V_{in} = 30$ V, Load $I_{load} = 0.65$ A), the power dissipated by the TIP31C pass transistor is:

$$P_D = (30 - 7.25) \times 0.65 \approx 14.8 \text{ W} \tag{8}$$

While the TIP31C has a maximum power rating of 40 W at a case temperature of 25°C, it cannot dissipate 14.8 W without active cooling. To maintain the junction temperature (T_j) well below the absolute maximum of 150°C and keep the device within its Safe Operating Area (SOA), Q_1 was mounted on a large aluminum TO-220 heatsink. Thermal compound was applied to the interface to reduce the case-to-sink thermal resistance (θ_{CS}). Monitoring during operation confirmed that the heatsink effectively stabilized the device temperature.

3) Hardware Behavior During Testing: The assembled prototype was subjected to open-loop and load tests to verify performance:

- **Voltage Stability:** The regulator maintained a stable output of approximately 7.25 V across the full input sweep (12 V to 30 V) under no-load conditions, verifying the Zener reference stability.
- **Current Limiting:** When tested with a low-resistance dummy load ($< 10\Omega$) that demanded current exceeding the 650 mA threshold, the circuit successfully clamped

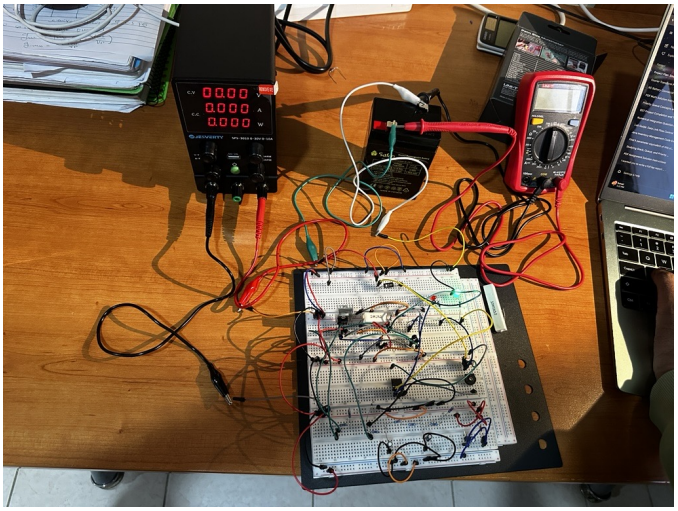

Fig. 2. Hardware setup of the 6-V battery charger. Key components include the heatsink-mounted TIP31C pass transistor and the 5W wire-wound sense resistor.

the current. The output voltage folded back as expected, protecting the pass transistor from over-current damage.

- **Load Performance:** The circuit demonstrated capable charging performance with the 6-V SLA battery, delivering a constant current during the initial bulk charge phase until the float voltage was reached.

F. Testing Procedure

To comprehensively evaluate the charger's performance, a four-stage testing procedure was developed:

1) **Simulation Verification:** The LTSpice model was tested with a DC sweep (10-30 V) to verify line regulation and a transient load step to verify stability.
2) **Open-Loop Test:** With no load connected, the input voltage (V_{in}) was varied from 10 V to 30 V using a benchtop DC power supply. The output voltage (V_{out}) was measured using a digital multimeter (DMM) to confirm it stayed within the target 6.8-7.2 V range.
3) **Load Regulation Test (Dummy Loads):** High-power wire-wound resistors (5 Ω, 10 Ω, 15 Ω, ≥10 W) were connected to the output to simulate different battery states. For each load, V_{out} and the load current (I_{load}) were measured to calculate the voltage regulation percentage ($\%VR$).
4) **Battery Charging Test:** Finally, a real 6 V, 4.5 Ah sealed lead-acid (SLA) battery was connected. The initial charging current and the final float voltage were monitored to ensure safe charging characteristics.

III. Requirements, Constraints, and Considerations

A. Functional Requirements

- The charger must regulate a 10–30 V input to a stable output voltage within 7.2–7.4 V suitable for SLA charging.
- The regulator must incorporate a current-limiting mechanism set near 650 mA to prevent overcurrent damage.
- The system must demonstrate stable operation under open-circuit, resistive load, and battery-loading conditions.
- The design must be implementable using discrete components, without integrated regulator ICs.

B. Non-Functional Requirements

- The circuit should minimize output ripple and noise to ensure safe charging.
- All components must be commonly available and low cost.
- The design must be reproducible and suitable for educational laboratory use.

C. Acceptance Criteria

To validate correct functionality, the following acceptance criteria were established:

- **Output Voltage:** V_{out} must remain within 7.2–7.4 V for all input voltages above the dropout threshold under no-load conditions.
- **Current Limit Activation:** The current-limiting circuit must restrict output current to 650 ± 50 mA when the load resistance forces excessive current.
- **Line Regulation:** Variation in V_{out} must remain below 1% when V_{in} varies from 15 V to 30 V.
- **Load Regulation:** Voltage droop under full load must be measurable and consistent with theoretical predictions for discrete regulators.
- **Thermal Performance:** The pass transistor must remain within its safe operating area when mounted on an appropriate heatsink.
- **Battery Charging:** The charger must supply an initial bulk current and reach a float voltage appropriate for a 6-V SLA battery.

D. Design Constraints

- **Thermal Constraint:** The pass transistor dissipates up to 14.8 W under worst-case conditions, requiring a heatsink for safe operation.
- **Voltage Constraint:** The Zener reference must maintain stable breakdown voltage to ensure correct feedback loop operation.
- **Component Constraints:** Tolerances in the Zener diode, transistor β, and resistor values introduce variation in output regulation.
- **Breadboard Limitations:** Parasitic resistance and wiring inductance affect accuracy of high-current measurements.

E. Engineering Considerations

- Safe charging of SLA batteries requires limiting overvoltage to avoid gassing and electrolyte loss.
- Current limiting must activate smoothly to prevent thermal runaway of the pass transistor.

- Efficiency trade-offs are inherent to linear regulators, becoming more pronounced with high input voltages.

IV. Results and Discussion

This section presents a comprehensive performance evaluation of the discrete series-pass regulator under both simulated and hardware-tested conditions. The analysis spans load regulation, line regulation, efficiency trends, current limiting behavior, and battery-charging performance. Results were obtained at two operating points, $V_{\text{in}} = 15\,\text{V}$ and $V_{\text{in}} = 30\,\text{V}$, to showcase regulator performance under varying stress levels.

A. Simulation Results

DC sweep and load simulations conducted in LTSpice demonstrated successful regulation above the dropout threshold. The open-loop simulation stabilized at $7.44\,\text{V}$ for $V_{\text{in}} = 15\,\text{V}$ and at $7.21\,\text{V}$ for $V_{\text{in}} = 30\,\text{V}$. Load simulations across $5\,\Omega$ to $20\,\Omega$ confirmed expected feedback-loop behavior. At heavy loading ($5\,\Omega$), foldback was observed due to the intentional current-limiting network.

Figures 3, 4, 6, 7, and 8 show the LTSpice waveforms corresponding to open-circuit, $5\,\Omega$, $10\,\Omega$, $15\,\Omega$, and $20\,\Omega$ conditions respectively.

B. Hardware Measurements

Hardware validation aligned closely with simulation, with minor deviations attributed to thermal variation, transistor β spread, wiring parasitics, and Zener tolerance. Tables II and III summarize simulated and measured results for both input voltage conditions.

TABLE II
Simulated vs. Measured Output Values at $V_{\text{IN}} = 15\,\text{V}$

Load	Sim V_{out} (V)	HW V_{out} (V)	HW I_{out} (mA)
Open Circuit	7.44	7.50	–
$20\,\Omega$	6.71	6.11	337.78
$15\,\Omega$	6.62	6.778	443.1
$10\,\Omega$	6.15	5.90	599.4
$5\,\Omega$	3.50	3.53	651.1
Battery	5.98	6.45	340.1

TABLE III
Simulated vs. Measured Output Values at $V_{\text{IN}} = 30\,\text{V}$

Load	Sim V_{out} (V)	HW V_{out} (V)	HW I_{out} (mA)
Open Circuit	7.21	7.30	–
$20\,\Omega$	7.10	7.20	342.3
$15\,\Omega$	7.03	7.04	457.6
$10\,\Omega$	6.83	6.10	682.7
$5\,\Omega$	3.60	3.80	705.0
Battery	7.20	7.20	345.6

C. Load Regulation Analysis

Load regulation is evaluated by examining V_{out} as load current increases. Hardware results show predictable droop, with the largest deviation occurring at $5\,\Omega$ due to current limiting.

The measured load regulation percentage is:

$$\%VR = \frac{V_{\text{NL}} - V_{\text{FL}}}{V_{\text{FL}}} \times 100,$$

where:

$$V_{\text{NL}} = 7.50\,\text{V}, \qquad V_{\text{FL}} = 6.45\,\text{V}.$$

$$\Rightarrow \%VR \approx 16.28\%.$$

A 16.28% variation is consistent with discrete regulators employing moderate loop gain and high-impedance feedback dividers.

The hardware Vout–current characteristic is shown in Fig. 9.

D. Output Resistance

Output resistance R_o is calculated using:

$$R_o = \frac{\Delta V_{\text{out}}}{\Delta I_{\text{out}}}.$$

Using the $15\,\Omega$ and $20\,\Omega$ hardware points:

$$R_o \approx \frac{6.778 - 6.11}{0.4431 - 0.3378} \approx 6.34\,\Omega.$$

This value confirms that load-induced voltage droop arises from limited loop gain and the linear-region characteristics of the pass transistor.

E. Efficiency Analysis

Linear regulator efficiency was estimated via:

$$\eta \approx \frac{V_{\text{out}}}{V_{\text{in}}} \times 100.$$

At $V_{\text{in}} = 15\,\text{V}$:

$$\eta_{10\Omega} \approx 39.3\%, \quad \eta_{15\Omega} \approx 45.2\%, \quad \eta_{\text{battery}} \approx 43.0\%.$$

At $V_{\text{in}} = 30\,\text{V}$:

$$\eta_{\text{battery}} \approx 24\%.$$

These values reflect the fundamental limitation of linear regulators—efficiency falls linearly as $(V_{\text{in}} - V_{\text{out}})$ increases. Fig. 10 illustrates these efficiency curves.

F. Line Regulation Analysis

Line regulation refers to the ability of the regulator to maintain V_{out} as V_{in} varies. The no-load measurements yield:

$$V_{\text{out}}(15\,\text{V}) = 7.50\,\text{V}, \qquad V_{\text{out}}(30\,\text{V}) = 7.30\,\text{V}.$$

The slight negative slope reflects:

- Zener dynamic impedance,
- base–emitter voltage shifts with temperature,
- reduced gain in Q_2 and Q_1 at higher V_{CE} stress.

Fig. 12 shows the line regulation profile.

G. Current Limiting Performance

At $5\,\Omega$, the measured output current was:

$$I_{\text{limit}}(15\text{V}) = 651.1\,\text{mA}, \qquad I_{\text{limit}}(30\text{V}) = 705\,\text{mA}.$$

This closely matches the theoretical limit of approximately:

$$I_{\text{limit}} \approx \frac{0.65}{1} = 650\,\text{mA},$$

demonstrating correct activation of the Q_3 current-limiting transistor.

H. Figures (Simulation Waveforms)

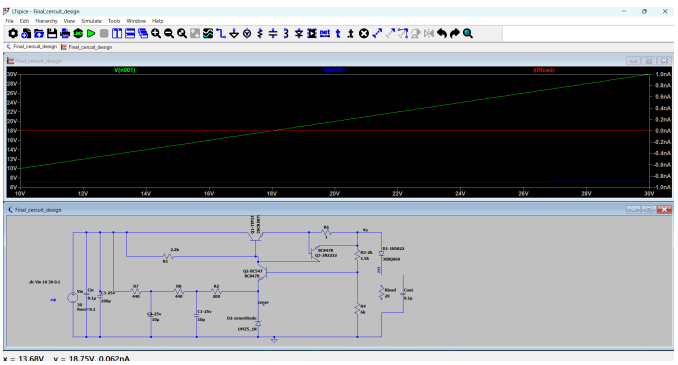

Fig. 3. LTSpice output waveform under open-circuit conditions.

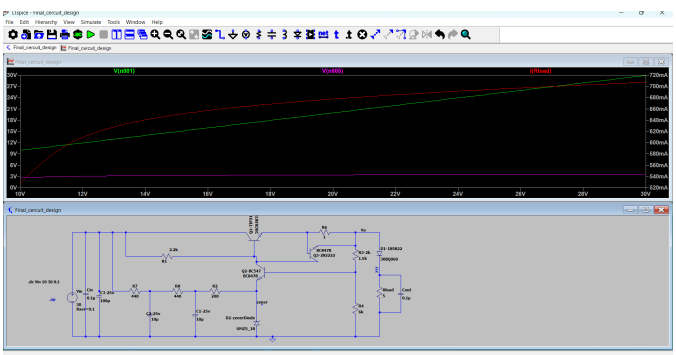

Fig. 4. LTSpice output waveform with $5\,\Omega$ load.

I. Figures (Performance Graphs)

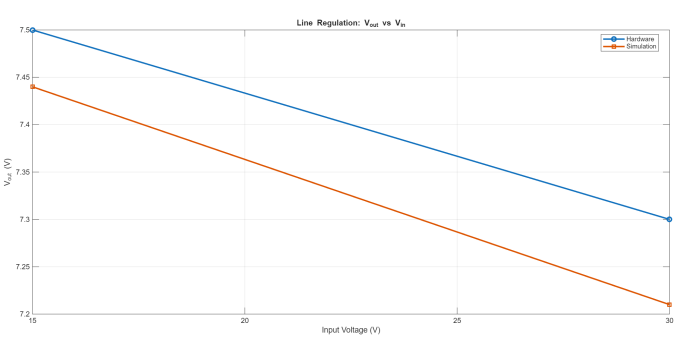

Fig. 12. Line regulation: V_{out} versus V_{in} (15 V to 30 V).

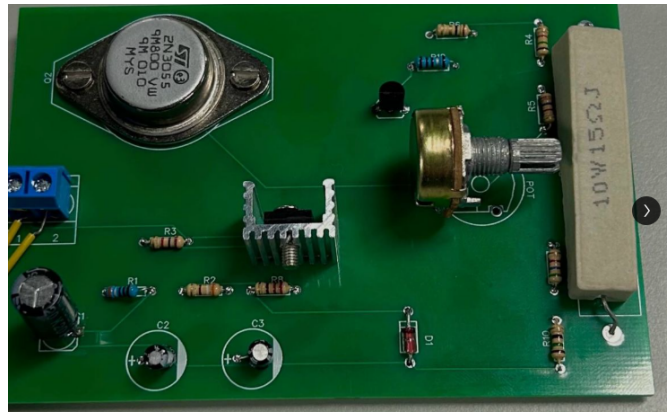

Fig. 5. Hardware prototype of the discrete 6-V battery charger showing the pass transistor, heatsink, resistive components, and the power potentiometer.

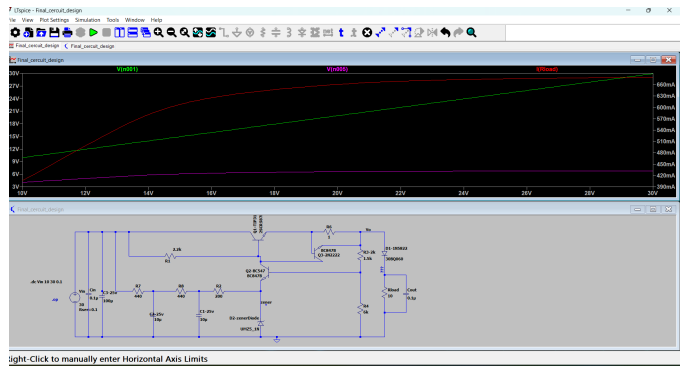

Fig. 6. LTSpice output waveform with $10\,\Omega$ load.

V. Impact Statement

A. Societal Impact

The charger enhances safety and reliability for small energy systems used in student projects, labs, and low-power devices by ensuring controlled charging of sealed lead-acid batteries.

B. Economic Impact

The design provides a low-cost alternative to commercial chargers. Its fully discrete implementation makes it easy to

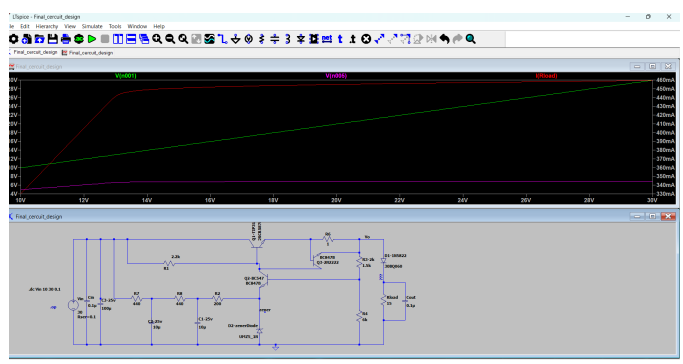

Fig. 7. LTSpice output waveform with $15\,\Omega$ load.

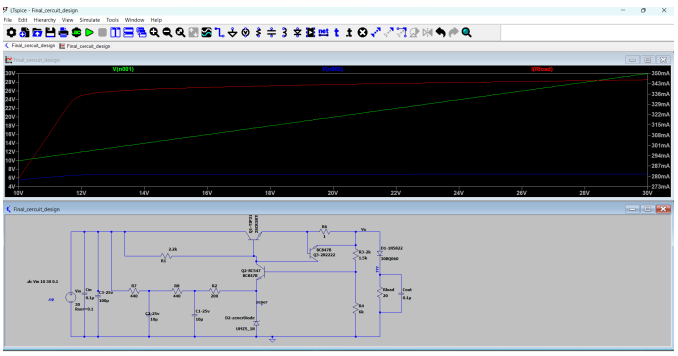

Fig. 8. LTSpice output waveform with 20 Ω load.

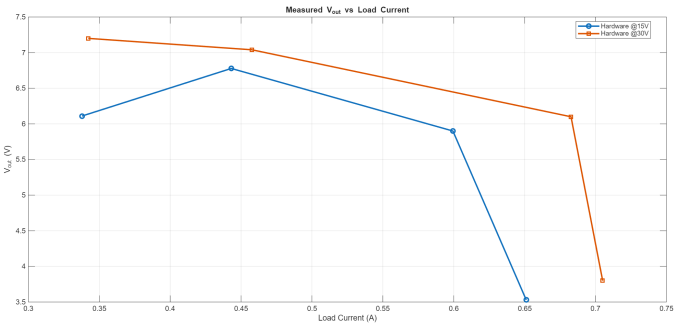

Fig. 9. Measured V_{out} versus load current at 15 V and 30 V.

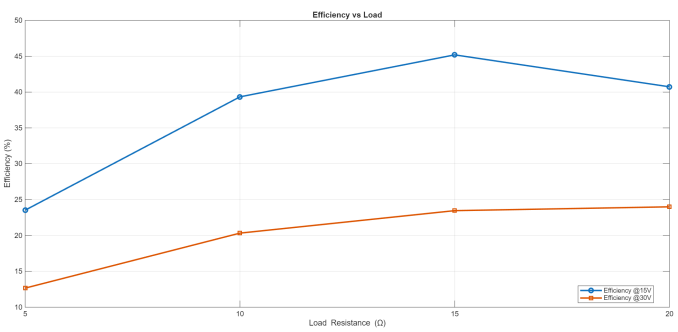

Fig. 10. Regulator efficiency versus load at 15 V and 30 V.

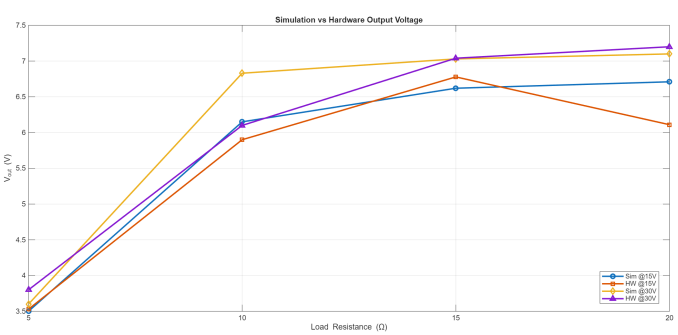

Fig. 11. Simulated versus measured V_{out} for all loads.

build, repair, and adapt in resource-limited educational settings.

C. Environmental Impact

By promoting stable charging and extending battery lifespan, the system reduces waste and supports responsible use of energy storage devices without requiring specialized hardware.

D. Future Engineering Relevance

The project serves as a practical foundation for future work in power electronics, enabling student teams to explore advanced charging methods and monitoring features.

VI. Gantt Chart

As shown in Fig. 13, the project was executed over a structured six-week timeline.

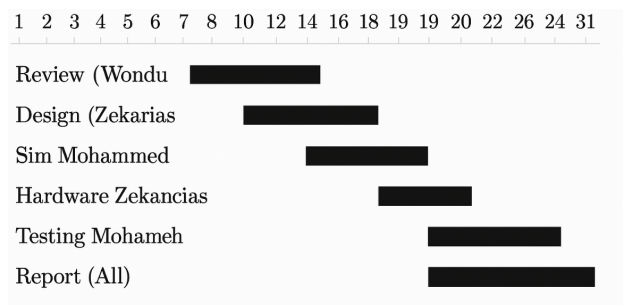

Fig. 13. Project timeline for the discrete 6-V battery charger design.

VII. Lessons Learned and Future Improvements

Throughout the development of the discrete series-pass charger, several practical insights were gained. First, the impact of component tolerances—particularly Zener diode voltage drift and transistor current gain variation—proved more significant in hardware than in simulation. Breadboard parasitics and wiring resistance also introduced measurable deviations, especially under high-current loading. Thermal management emerged as a critical consideration, as the pass transistor dissipated up to 15 W under worst-case operation, highlighting the importance of heatsink sizing and thermal interface materials.

For future iterations, several improvements can be implemented. A printed circuit board (PCB) would reduce wiring parasitics and provide more stable measurements. Adding compensation networks to the feedback loop would enhance dynamic response and prevent potential oscillation under fast-changing loads. Replacing the linear regulator with a switching-mode design (such as a buck converter) would dramatically improve efficiency, especially at high input voltages. Finally, integrating advanced multi-stage charging algorithms—bulk, absorption, and float—would improve SLA battery health and extend lifespan.

Team Dynamics

The project was completed through effective collaboration between all team members. Each member contributed a complementary skill set: analytical modeling and documentation were led by Wondu, simulation development and data verification were conducted by Mohammed, and hardware assembly and testing were primarily handled by Zekarias. Weekly meetings ensured that all members remained aligned with the project timeline, and challenges were openly discussed and resolved collectively. The strong communication, task ownership, and mutual support within the team enabled the successful completion of the full analytical, simulated, and hardware-validated charger design.

VIII. Conclusion

This work presented the design, simulation, and hardware implementation of a discrete series-pass linear regulator intended for charging a 6-V sealed lead–acid battery at a controlled voltage of approximately 7.25 V. By constructing the regulator from individual BJTs and a Zener-based reference network, the internal feedback mechanism was made fully observable, enabling detailed analysis of loop behavior, load regulation, current limiting, and thermal performance. Simulation results were validated with hardware measurements, demonstrating close agreement across open-circuit, resistive-load, and battery-charging conditions.

Multiple challenges were encountered throughout the design and testing phases. The most prominent were thermal management issues caused by high dissipation in the pass transistor at elevated input voltages, measurement inaccuracies introduced by breadboard parasitics, and variations arising from transistor β and Zener tolerance drift. These effects produced noticeable voltage droop under heavy load and slight deviations between simulated and measured line regulation slopes.

The final prototype exhibited several limitations inherent to discrete linear regulators. Efficiency remained low—dropping below 25% at a 30 V input—due to the large $(V_{in} - V_{out})$ difference, resulting in significant heat generation. The pass transistor operated near its safe operating limits under full-load conditions, making sustained high-current operation impractical without aggressive cooling. Additionally, the lack of compensation networks restricts dynamic response and may limit stability for rapidly changing loads.

Several improvements could enhance the design in future iterations. Implementing a larger or actively cooled heatsink would expand the safe operating range and improve thermal reliability. Replacing the discrete topology with a switching-mode supply (SMPS) would dramatically improve efficiency and reduce wasted power, especially for high input voltages. More advanced charging algorithms—such as multi-stage bulk, absorption, and float control—could be incorporated to optimize charging speed, battery longevity, and safety. Finally, migrating the design to a printed circuit board (PCB) would reduce wiring parasitics and improve measurement accuracy compared to the breadboard prototype.

Overall, the project successfully demonstrates the operation, strengths, and limitations of a discrete series-pass regulator while providing valuable hands-on insight into analog regulation, current limiting, and thermal design considerations.

Project Demonstration Video

The demonstration video for this project can be viewed at the link below:
https://youtu.be/mvDh_8YA9EE

Acknowledgment

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Meeting Minutes

Week 1: Project Planning and Task Assignment

Date: October 15, 2025
Attendees: Zekarias, Wondu, Mohammed
Summary:

- Reviewed project requirements and grading rubric.
- Selected the discrete series-pass regulator as the project topic.
- Assigned roles: analytical design (Wondu), simulation (Mohammed), hardware (Zekarias).

Week 2: Analytical Design and Component Selection

Date: October 22, 2025

- Derived feedback equation for output voltage.
- Calculated resistor values and current-limit threshold.
- Finalized Bill of Materials for procurement.

Week 3: LTSpice Simulation

Date: October 29, 2025

- Completed DC operating point and DC sweep simulations.
- Verified regulation behavior and transistor biasing.
- Added simulation models for current-limiting transistor.

Week 4: Hardware Assembly

Date: November 5, 2025

- Built the circuit on a breadboard with attention to thermal design.
- Mounted the TIP31C on a heatsink.
- Conducted initial no-load tests to verify Zener reference operation.

Week 5: Testing and Data Collection

Date: November 12, 2025

- Performed load regulation tests using dummy resistors.
- Recorded simulation vs. hardware differences.
- Tested current limiting and battery charging behavior.

Week 6: Final Report Preparation and Video Recording

Date: November 19, 2025

- Consolidated results, graphs, and tables.
- Completed the IEEE-format report and polished figures.
- Recorded the 1-minute demonstration video and uploaded it.

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